

FSK176: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2015

Chapter 5 Laws of Motion / Hoofstuk 5 Bewegingswette

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P5.22 (a) 181° counter-clockwise from the positive x -axis (b) 11.2 kg (c) 37.5 m/s (d) $\vec{v}_f = (-37.5\hat{i} - 0.893\hat{j})$ m/s

P5.29 (a) 7.0 m/s^2 horizontally to the right (b) 21 N (c) 14 N horizontally to the right

P5.34 Proof

P5.49 (a) 2.20 m/s^2 (b) 27.4 N

P5.64 (a) FBD's (b) 2.31 m/s^2 , down for m_1 , left for m_2 , and up for m_3 (c) $T_{\text{left}} = 30 \text{ N}$, $T_{\text{right}} = 24.2 \text{ N}$ (d) Explanation

P5.66 (a) $P_{\text{max}} = 48.6 \text{ N}$ and $P_{\text{min}} = 31.7 \text{ N}$ (b) Explanation (c) The block can not slide up the wall. If $P < 62.7 \text{ N}$, it will slide down the wall.

P5.88 Static friction coefficient is $\mu_s = m_s$

(a) $M = 3m \sin \theta$ (b) $T_1 = 2mg \sin \theta$, $T_2 = 3mg \sin \theta$ (c) $a = \frac{g \sin \theta}{1+2 \sin \theta}$,

$T_1 = 4mg \sin \theta \left(\frac{1+\sin \theta}{1+2 \sin \theta} \right)$, $T_2 = 6mg \sin \theta \left(\frac{1+\sin \theta}{1+2 \sin \theta} \right)$ (e) $M_{\text{max}} = 3m(\sin \theta + \mu_s \cos \theta)$ (f) $M_{\text{min}} = 3m(\sin \theta - \mu_s \cos \theta)$ (g) $T_{2,\text{max}} - T_{2,\text{min}} = M_{\text{max}}g - M_{\text{min}}g = 6\mu_s mg \cos \theta$

P5.89 (a) Proof (b) If $\tan \theta < \frac{1}{\mu_s}$ is not met, no value of P can move the crate.